Conditional zero order at an actual projective branch

Formal proof packet

27 October 2026

This packet records a conditional local-geometry proof, not a target experiment or cost comparison. The editable diagram, gate CSV, Lean source, and compiler log are beside this PDF.

Checked statement

Let a be a rational simple root of the quintic and R the structure-sheaf stalk of the actual weighted-Proj curve at that branch. The previously checked local geometry makes R a DVR, with $\operatorname{ord}(y) = 1$ and $y^2 = (x - a)q$ for a unit q. The new theorem proves that if $N = (x - a)t$ and B is a unit in R, then

$$\operatorname{ord}(N - By) = 1.$$

The proof transports the generic branch zero-order lemma into this actual projective stalk. The underlying DVR criterion is discussed in the Stacks Project, Lemma 10.119.7.

Checked actual stalk: R is a DVR and $\operatorname{ord}(y) = 1$

Explicit input: $N = (x - a)t$, and B is a unit

$\Downarrow$ checked local valuation algebra

$\operatorname{ord}(N - By) = 1$

Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero. The theorem axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Lean source SHA-256:

`49005acc564b41341b37296d42de90c48e56ac4f4a4ec4860d7278fc8bc00cca.`

The compiled Rust law has not been shown to satisfy the numerator factorization or unit hypothesis at every branch. Nonbranch multiplicities, the global principal divisor, the Pic^0 class equation, and universal runtime semantics remain open. No frozen 1,260-target row or matched-rho ratio changes.